

Assignment 7

TCP Basics

- Start up your web browser. Go the <http://gaia.cs.umass.edu/wireshark-labs/alice.txt>
- Save a copy of *Alice in Wonderland*. Store this file somewhere on your computer. (right click anywhere on the page >> click save as)
- Next go to:
<http://gaia.cs.umass.edu/wireshark-labs/TCP-wireshark-file1.html>
- Use the Browse button in the form to enter the name of the file (alice.txt)
Don't yet press the "Upload alice.txt file" button.
- Now start up Wireshark and begin packet capture.
- Returning to your browser, press the "Upload alice.txt file" button to upload the file to the gaia.cs.umass.edu server.
- Once the file has been uploaded, a short congratulations message will be displayed in your browser window.
- Stop the packets capturing and apply "TCP" filter.



- 1- $\text{Syn} = 0$
- 2- $\text{Ack} = x + 1$ and $\text{syc} = y$
- 3- $\text{Ack} = y + 1$

1. What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu?

```

112.7.110698000    172.22.34.109    128.119.245.12    TCP    66 42525 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
+ Internet Protocol Version 4, Src: 172.22.34.109, Dst: 128.119.245.12
+ Transmission Control Protocol, Src Port: 42525, Dst Port: 80, Seq: 0, Len: 0
  Source Port: 42525
  Destination Port: 80
  [Stream index: 19]
  [Stream Packet Number: 1]
  [Conversation completeness: Incomplete, ESTABLISHED (7)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 1589191832
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 0
  Acknowledgment number (raw): 0
  1000 .... = Header Length: 32 bytes (8)
  Flags: 0x002 (SYN)
  Window: 64240
  [Calculated window size: 64240]
  Checksum: 0x0659 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (12 bytes) Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), SACK permitted

```

sequence number is: 0

2. What is it in the segment that identifies the segment as a SYN segment?

```

112.7.110698000    172.22.34.109    128.119.245.12    TCP    66 42525 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
+ Flags: 0x002 (SYN)
  000. .... = Reserved: Not set
  ...0 .... = Accurate ECN: Not set
  ....0... = Congestion Window Reduced: Not set
  ....0... = ECN-Echo: Not set
  ....0... = Urgent: Not set
  ....0... = Acknowledgment: Not set
  ....0... = Push: Not set
  ....0... = Reset: Not set
  ....0... = Syn: Set
  ....0... = Fin: Not set
  [TCP Flags: .....S.]

```

In transmission control protocol go to { flags : (syn)} everything is not set only (syn) is set

3. What is the sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN?

```

217.8.264814800    128.119.245.12    172.22.34.109    TCP    66 80 → 42525 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128

```

Syn number is: 0

4. What is the value of the ACKnowledgement field in the SYNACK segment?

```

217.8.264814800    128.119.245.12    172.22.34.109    TCP    66 80 → 42525 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128

```

ack number is: 1

5. What is the IP address and TCP port number used by the computer (source) that is transferring the file to gaia.cs.umass.edu?

```

287.8.354178200    172.22.34.109    128.119.245.12    HTTP    1125 POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
307.8.524271300    128.119.245.12    172.22.34.109    HTTP    834 HTTP/1.1 200 OK (text/html)

```

IP address: 172.22.34.109

```

+ Transmission Control Protocol, Src Port: 42504, Dst Port: 80, Seq: 151958, Ack: 1, Len: 1071
  Source Port: 42504
  Destination Port: 80
  [Stream index: 21]
  [Stream Packet Number: 149]
  [Conversation completeness: Incomplete (28)]
  [TCP Segment Len: 1071]
  Sequence Number: 151958 (relative sequence number)
  Sequence Number (raw): 3486148072
  [Next Sequence Number: 153029 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 1923011659
  0101 .... = Header Length: 20 bytes (5)

```

Go to transmission control protocol the search for source port: 42504

6. What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?

287	8.354178200	172.22.34.109	128.119.245.12	HTTP	1125 POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
307	8.524271300	128.119.245.12	172.22.34.109	HTTP	834 HTTP/1.1 200 OK (text/html)

IP address: 128.119.245.12

```
- Transmission Control Protocol, Src Port: 42504, Dst Port: 80, Seq: 151958, Ack: 1, Len: 1071
  Source Port: 42504
  Destination Port: 80
  [Stream index: 21]
  [Stream Packet Number: 149]
  [Conversation completeness: Incomplete (28)]
  [TCP Segment Len: 1071]
  Sequence Number: 151958 (relative sequence number)
  Sequence Number (raw): 3486148072
  [Next Sequence Number: 153029 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 1923011659
  0101 .... = Header Length: 20 bytes (5)
```

Go to transmission control protocol the search for Destination port: 80

Transmission control protocol = TCP

- 1- Reliable
- 2- No data lose
- 3- No interruption
- 3 way – handshake

Server <----- (SYN) -----> browser (capture) syn = X

Server <----- (SYN), (ACK) -----> browser ACK = X + 1 --- SYN = Y

Server <----- (ACK) -----> browser ACK = Y + 1

{ Connection Establisshed }